

Green Metals

Battery Metals Watch: The rationing phase



Too early to call the bottom. Despite the significant downside in battery metals prices this year, with nickel, lithium and cobalt prices down 60%, 70% and 60% from last year's peak, respectively, we believe it is too early to call an end to these respective bear markets. This year, we believe a rise in supply combined with a shift lower in retail EV volume growth in China (+2.7Mil units in 2022 vs +1.5Mil units in 2023E) has weighed on prices. However, in our view, the oversupply led low-cycle pricing is yet to arrive and the size of upcoming surpluses has not yet been fully realised in prices. The lithium market is largely balanced this year, but we continue to see a massive surplus (202kt vs 245kt previously) over next year. For nickel, we think increasing class 1 supply and rising refined metal supply will continue to drive prices lower until supply responses emerge. Lastly, the cobalt metal price has risen 11% since mid-Sep due to reports of strategic stockpiling but look overpriced for the continued glut (10kt and 15kt surplus 2023-24E). Whilst the supply rationing phase that lies ahead in these commodities will likely generate more volatility, we remain bearish on the battery metals complex with our 12m price target for lithium carbonate at \$15,000/t, nickel at \$16,000/t and cobalt at \$28,000/t (vs 25,000/t previously).

A new era of EV demand awaits. With the phase-out of national subsidies in China and reduction of subsidies in some European countries, the EV market is now becoming increasingly dependent on consumer-led adoption. Compared to the West, we view China to be closest to a consumer-led EV adoption as Chinese EVs are competitively priced against ICEs in the local market. China's retail EV passenger sales are estimated to rise to 8.9Mil units in 2024 on our auto analysts' projections. In Europe, challenging economic conditions, roll back of incentives and lower price competitiveness of EVs (vs ICE) have raised concerns over EV uptake near term; however, with support from new model launches and stringent CO2 targets in 2025. Lastly, the US EV market will be increasingly supported by the IRA credits, economies of scale and charging agreements amongst major OEMs though also face risk from high interest rates. Overall, we expect global EV sales (PHEV+BEV) of 13.2Mil units in 2023 and 16.8Mil units in 2024 based on our auto analysts' projections. Whilst near-term upside surprises to demand look limited, our long-term demand estimates have risen on rising average US battery size, higher EV sales and energy storage.

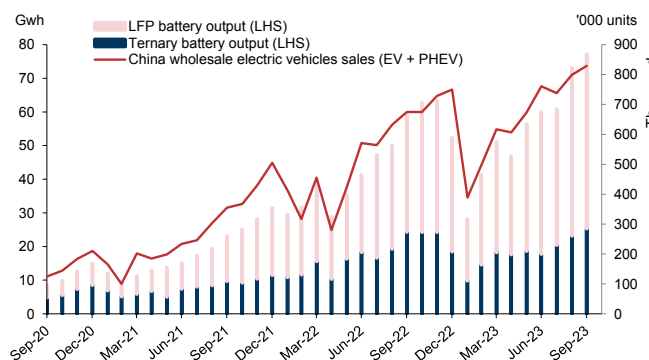
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All eyes on Chinese supply. With respective supply ramp-ups set to more than meet demand and prices necessarily correcting into cost curves, the key question is now which price levels generate supply adjustments. For lithium, the focus is on China, the higher cost production location, where on aggregate 2023-24 supply remains in line with our expectations. Moreover, the non-integrated cost of production using purchased spodumene is expected to fall to a cost of \$14k/t LCE (vs. \$27k/t LCE presently) on a lagged correction in spodumene prices by mid-next year, on our estimates. Furthermore, our China team has maintained their estimate of total cost for the integrated lepidolite between \$11-14k/t LCE, which is substantially below current price levels. We continue to see China lithium carbonate prices declining to \$15,000/t LCE on a 12m horizon, but lower our Asia CJK CIF Li hydroxide forecast to \$16,500/t (from \$19,000/t previously). The main risk to a low-cycle in lithium pricing in 2H2024, in our view, comes from delays to Chinese expansions from limitations to land quota required for tailing management or slower spodumene ramp-ups. In such a scenario, we expect China lithium carbonate prices to find support around RMB 100-120k/t LCE (vs RMB 80k/t LCE GSe in 2H2024).

The rationing phase

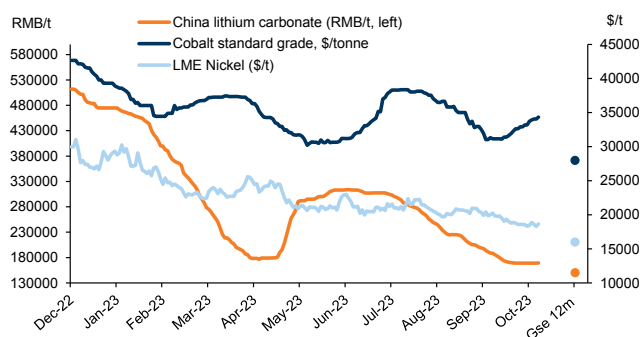
1. Too early to call the bottom. Despite the significant downside in battery metals prices this year, with nickel, lithium and cobalt prices down 60%, 70% and 60% from last year's peak, respectively, we believe it is too early to call an end to these respective bear markets. This year, we believe a rise in supply combined with a shift lower in retail EV volume growth in China (+2.7Mil units in 2022 vs +1.5Mil units in 2023E) has weighed on prices. However, in our view, the oversupply led low-cycle pricing is yet to arrive and the size of upcoming surpluses has not yet been fully realised in prices. The lithium market is largely balanced this year, but we continue to see a massive surplus build (202kt vs 245kt previously) over next year ([Exhibit 3](#)). For nickel, we think increasing class 1 supply and rising refined metal supply will continue to drive prices lower until supply responses emerge. Lastly, cobalt metal price has risen 11% since mid-Sep due to reports of strategic stockpiling but look overpriced for the continued glut (10kt and 15kt surplus 2023-24E). Whilst the supply rationing phase that lies ahead in these commodities will likely generate more volatility, we remain bearish on the battery metals complex with our 12m price target for lithium carbonate at \$15,000/t, nickel at \$16,000/t and cobalt at \$28,000/t (vs 25,000/t previously) ([Exhibit 2](#)).

Exhibit 1: Whilst demand has continued to rise, incremental sales volumes and battery output has moderated



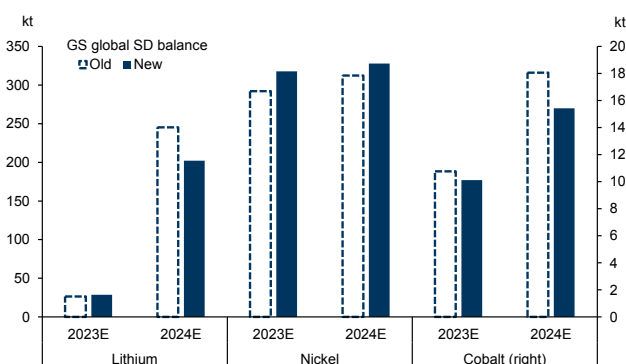
Source: Goldman Sachs Global Investment Research, CAAM, CPCA, Wind

Exhibit 2: The YTD correction in battery prices has been driven by moderation in demand; we expect a further fall on oversupply in H2



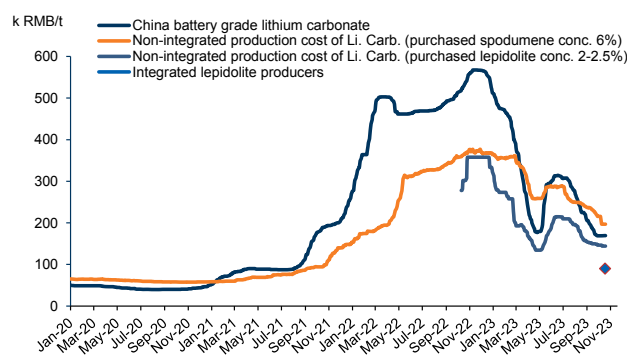
Source: Goldman Sachs Global Investment Research, Fastmarkets, SMM

Exhibit 3: Upcoming supply growth will likely push battery metals into bigger surplus over 2024



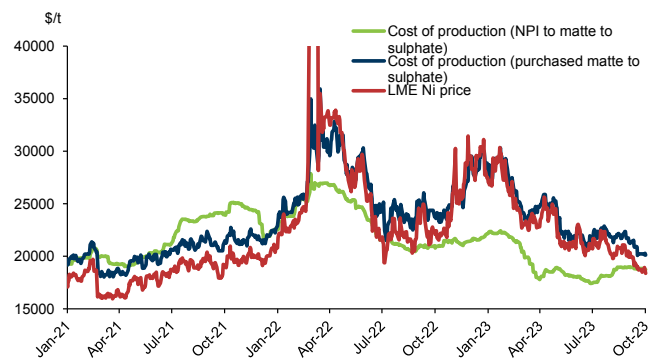
Source: Goldman Sachs Global Investment Research

Exhibit 4: Lithium prices are pushing non-integrated capacity into loss-making territory, but still above integrated cost of production



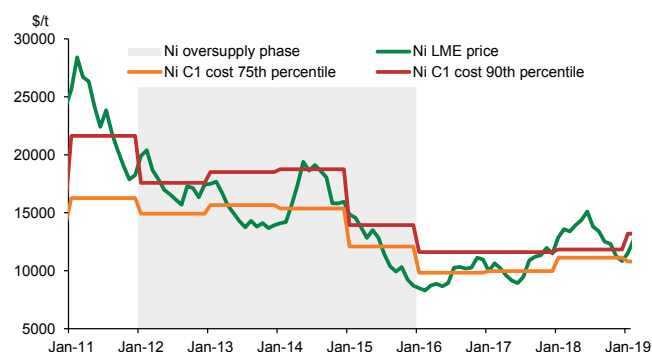
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 5: For nickel, we have now entered the meat of the cost curve with prices around \$18,500/t and margins are coming under pressure...



Source: Goldman Sachs Global Investment Research, SMM

Exhibit 6: ...however prices will need to dip below the 90th percentile – like in the mid 2010s – to restrain the expected glut in the coming years

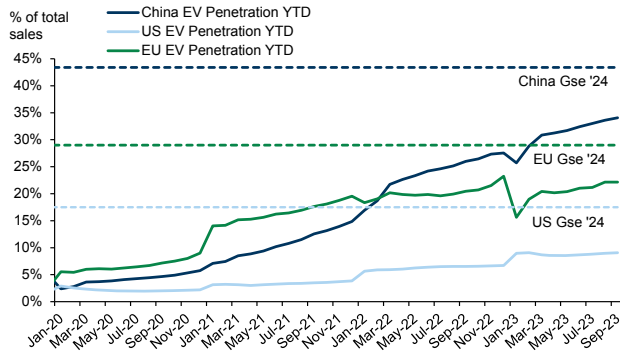


Source: Goldman Sachs Global Investment Research, Woodmac

2. A new era of EV demand awaits. With the phase-out of national subsidies in China and reduction of subsidies in some European countries, the EV market is now becoming increasingly dependent on consumer-led adoption. Compared to the West, we view China to be closest to a consumer-led EV adoption as Chinese EVs are competitively priced against ICEs in the local market. China's passenger retail EV sales are estimated to rise to 8.9Mil units in 2024 on our auto analysts' projections. In Europe, challenging economic conditions, roll back of incentives and lower price competitiveness of EVs (vs ICE) have raised concerns over EV uptake near term; however, with support from new model launches and stringent CO2 targets in 2025 our autos team expects EU sales in 2024 at 3.7Mil units. Lastly, the US EV market (1.7Mil in 2023) will be increasingly supported by the IRA credits, economies of scale and charging agreements amongst major OEMs though also face risk from high interest rates. Overall, we expect global EV sales (PHEV+BEV) of 13.2Mil units in 2023 (+3.7Mil units y/y) and 16.8Mil units in 2024 (+3.6Mil units y/y) based on our auto analysts' projections. Whilst EV battery demand upside is looking more complicated near term, our long-term demand estimates have risen on bigger US battery size and higher EV sales over 2025-30E (Exhibit 11). Furthermore, energy storage acceleration has continued to boost long-term battery

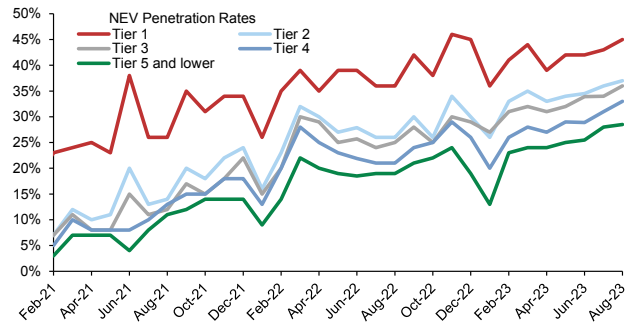
demand, in our view. We expect ESS battery demand now to represent 11% of total battery demand (vs 9% a year ago) due to rising demand from renewable installation, particularly in China (Exhibit 9).

Exhibit 7: The west is still far behind China on the EV adoption curve, leaving room for upside in demand



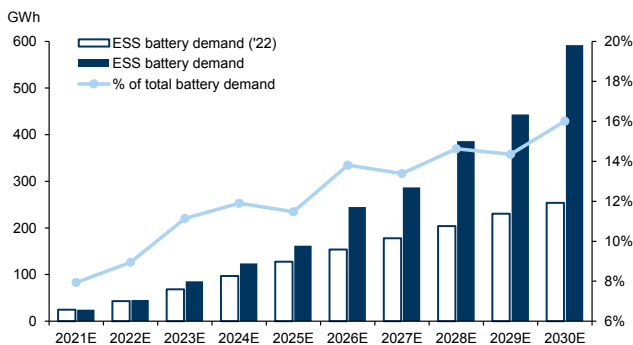
Source: Goldman Sachs Global Investment Research, Wind, IHS, Haver Analytics

Exhibit 8: In China, EV penetration in tier 2 cities is lagging and requires more infrastructure investment as well as subsidy support



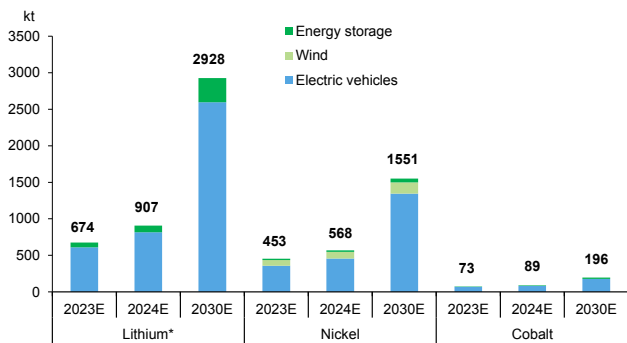
Source: Goldman Sachs Global Investment Research, CPCA

Exhibit 9: Long-term estimates of ESS battery demand have continued to increase on higher renewables installations in China



Source: Goldman Sachs Global Investment Research, BNEF, IEA

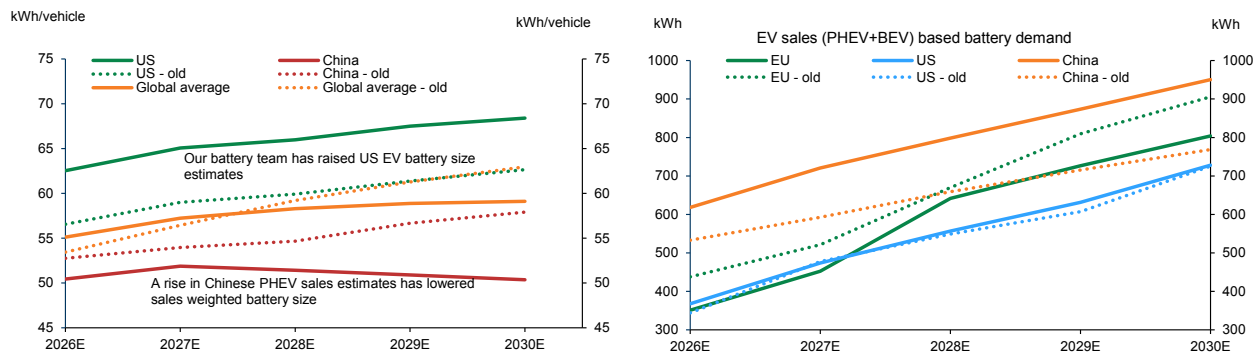
Exhibit 10: Battery metals long-term demand outlook looks resilient despite near-term challenges
Battery metals green demand



*Lithium is in kt LCE

Source: Goldman Sachs Global Investment Research

Exhibit 11: Long-term outlook for EV battery demand stays resilient as we raise our estimates for US battery size and China EV sales

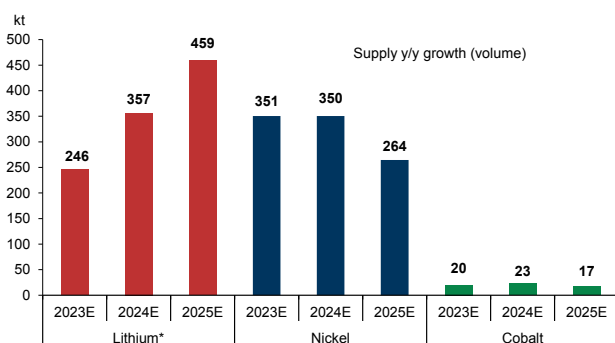


Source: Goldman Sachs Global Investment Research

3. Supply set to outpace demand. Despite some supply rationing and delay in ramp-ups due to land-quota and environmental issues, Chinese lithium chemical production has grown +33%/y/y YTD. Spodumene imports have risen by 47%/y/y year-to-August, and on our analysts' estimate spodumene projects continue to ramp up in-line with slight delays on certain project (detailed analysis below). In Latam, Sigma Lithium expects Phases 2 and 3 of its project to reach full capacity in the coming years - which should take yearly production to 104ktpa LCE, where our mining team conservatively forecast the project's production at 93ktpa LCE in 2027. Our China materials team continue to expect supply additions in the coming years from Chinese owned (fully/partly) African assets (Arcadia, Bikita, Kamativi, Sabi Star and DMCC) on successful project ramp-ups, but note that risks from policy shifts and logistics remain. For nickel, we see the main additions in class 1 supply coming from the ramp-up of chemical production through NPI to matte conversion and HPAL (GEM, CNGR, Jinchuan and Huayou Cobalt), alongside new metal capacities in China. We expect Chinese refined assets to add c.250kty of supply in the form of metal and nickel sulphate in 2023-24 (Exhibit 36). Furthermore, while not in our base case, we note reports of 80kty of Indonesian refined metal capacity under construction, which should further add to growing class 1 supply. Finally, for cobalt, growing mixed hydroxide precipitate production – which holds both cobalt and nickel via the HPAL process - is contributing to an increasing cobalt intermediate supply and with the ramp-up of refining capacity in China will increase cobalt metal supply in the coming 12 months.

Exhibit 12: Supply continues to surge for battery metals complex

Disruption adjusted supply refined supply growth

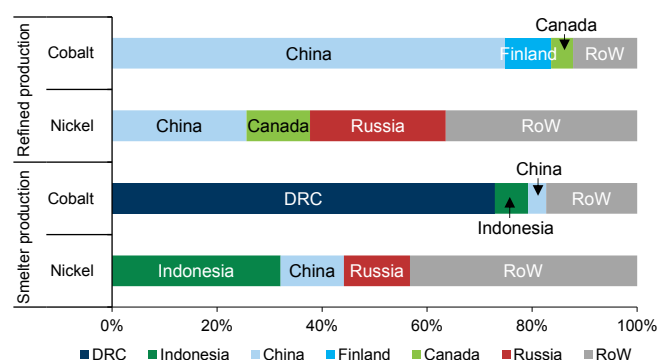


*Lithium is in kt LCE

Source: Goldman Sachs Global Investment Research, Company data, Woodmac, BNEF

Exhibit 13: Both class 1 nickel and cobalt prices are exposed to supply shock volatility due to high concentration

Based on 2023 GS estimates



Source: Goldman Sachs Global Investment Research, Woodmac

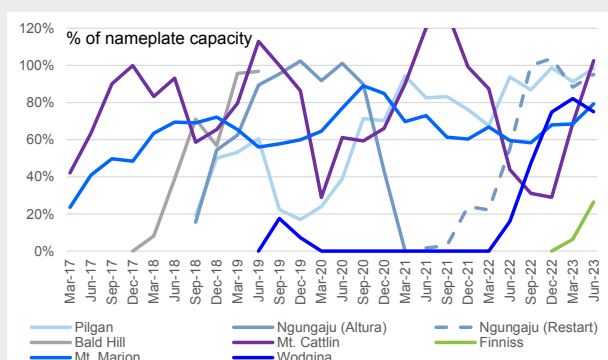
Australian Lithium Supply Update (Hugo Nicolaci*)

A common counterargument to the rapidly increasing spodumene/lithium supply view has been that these projects are more difficult than expected and take a longer time to ramp up, with the implication that these projects are not adding meaningful volumes to the market in the interim (though most producers still expect their own projects to ramp up on schedule). While historically it has been partially true there have been difficulties reaching nameplate spodumene capacity, **our analysis suggests that new projects have, on average, reached 60-80% of nameplate capacity within 9-12 months of start up**, where ~55-60% recovery rates have then since increased as projects target a lower spodumene product grade.

Furthermore, feedback from these existing producers (including at recent site visits) is that these types of projects are getting easier with learnings from their own experiences and a growing pool of expertise in the industry. This is consistent with feedback from both developers (including two projects in construction), and explorers (with a further six Australian spodumene projects in study), that learnings from previous projects have improved flowsheet designs and should result in a more predictable ramp up, though producers expect to retain an advantage on greater experience.

Exhibit 14: Australian spodumene projects have had varying ramp up rates...

Australian hard rock project production as % of SC6 nameplate

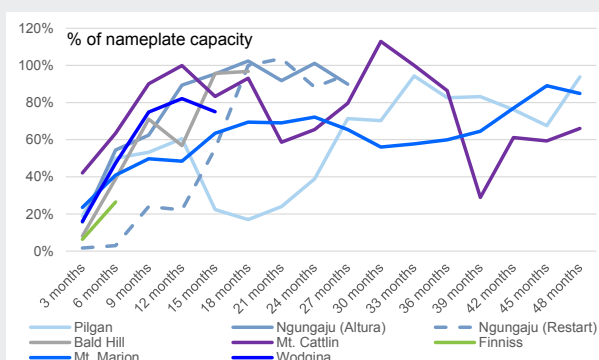


Performance against SC6 nameplate capacity will vary with ore feed grade and recovery; Utilisations turned down during pricing downturn in 2019/20; Wodgina taken on Trains 1&2, Train 3 taken from mid-2023; Mt. Marion expansion taken from mid-2023; PLS plant splits GSe from FY23.

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: ...though, on average, reached 60-80% of nameplate capacity within 9-12 months of start up...

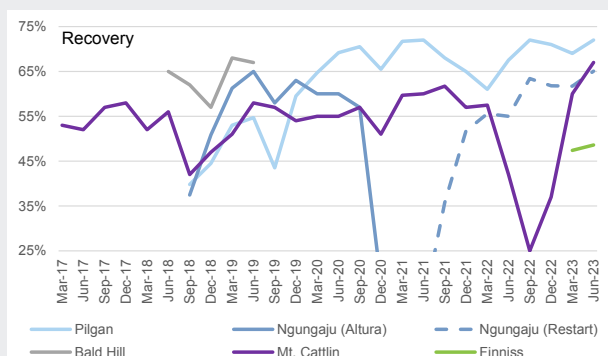
Australian hard rock production ramp up as % of SC6 nameplate (adjusted to first quarter of production)



Wodgina taken from Jun-22 (rather than brief production in late 2019) on Trains 1&2; Projects may go beyond adjusted ramp up chart timeframe.

Source: Company data, Goldman Sachs Global Investment Research

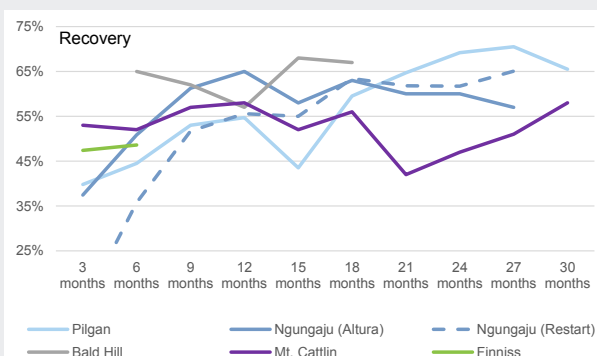
Exhibit 16: ...with recoveries hitting ~55-60% in the first year...
Australian hard rock project recovery (%)



Recoveries may vary with target spodumene product grade; Quarterly recovery data for Wodgina/Mt. Marion, ramp up data for Greenbushes CGP1&2, and data for Bald Hill restart not reported.

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 17: ...and since increasing as projects target a lower spodumene product grade
Australian hard rock project recovery (%; adjusted to first quarter of production)



Projects may go beyond adjusted ramp up chart timeframe.

Source: Company data, Goldman Sachs Global Investment Research

Concerns have also remained regarding access to **project funding**, however we expect Liontown's recent debt package from commercial/government lenders as part of its broader funding strategy may de-risk commercial/bank funding for future spodumene/lithium projects. The development of regional lithium districts may also de-risk upfront project funding where processing of third party feed could support development of smaller/more marginal assets (which would no longer require their own process plant/associated upfront capex spend). Separately, we note that recent downgrades to industry body lithium price forecasts may be increasingly negative for the NPV of any projects with upcoming studies.

With regard to **downstream partnering processes** (i.e. PLS seeking further downstream partnerships, LTR's recent partnership with Sumitomo, etc.), while the balance of capacity additions is now more favorable to the upstream than downstream converters vs. the last cycle (with some spodumene producers highlighting the potential for more negotiating power), the potential scale of unallocated/spot spodumene volumes coming to market could reduce the likelihood of upstream producers getting significantly more generous terms (such as free carry on their share of downstream plant capex), in our view, though partnerships to make the most of government incentives remain likely.

Projects supply update

A summary of the state of the key Australian upstream producers/projects.

1. Talison's (Tianqi/Albemarle/IGO) **Greenbushes** (1.4-1.5Mt spodumene concentrate production guidance in FY24, from ~1.5Mt in FY22) continues to support growing production, where chemical grade plants 3 & 4 potentially add an incremental 520ktpa of capacity each from mid-late decade.
2. Pilbara Minerals' **Pilgangoora** targets growing production to 600-630kt (5.7%) in FY24, ramping up to 680ktpa from 4QCY23 with a 100ktpa expansion coming online, and with a further expansion to >1Mtpa by FY26 following FID earlier this year (with additional volumes currently unallocated). Studies for the potential to go beyond P1000 are ongoing, with study results expected potentially some time in 1H CY24.

3. At **Wodgina** (Mineral Resources/Albemarle), Train 3 (additional 250ktpa) is expected to be running in early CY24, as highlighted by the recent site visit, with studies still progressing on a 4th train (MIN still expect commissioning by June 2025).
4. **Mt. Marion** (Mineral Resources/Ganfeng) processing plant upgrades are complete, and MIN has reaffirmed its medium-term spodumene production target of 900ktpa (600ktpa SC6); however, we believe mining rates will be the biggest challenge for MIN and assume they reach 780ktpa (490ktpa SC6) run-rate by the Sep Q 2024.
5. Allkem's **Mt Cattlin** spodumene project expects to have rebounding volumes into FY24, with AKE guiding 210-230kt, and the Stage 4 Expansion project technical report highlighting a LOM to Sep-27. At **James Bay**, AKE note discussions/study reviews (IBA/COMEX) are in the final stages and expect ~19 months of construction once provincial authorisation is obtained, though we remain more conservative on timing where we expect first material movement in mid-CY24 and first spodumene production in FY26, conservatively ramping up to ~320ktpa by FY28. On brine assets, AKE guide to Olaroz FY24 production of 22-26kt lithium carbonate for Stage 1 and Stage 2 following first production from Olaroz Stage 2 in mid-July (12-18 month ramp up expected), taking Olaroz to ~40ktpa of lithium carbonate by mid-decade (before enhanced recovery works), while a Stage 3 expansion (not in GSe) could take Olaroz to ~100ktpa LCE (possibly including Cauchari). At **Sal de Vida** the two-stage 45ktpa carbonate project is under construction and targeting substantial mechanical completion, pre-commissioning and commissioning activities in 1HCY25, with first production from the 15ktpa Stage 1 to follow in 2HCY25 (ramp up 12 months), with Stage 2 first production targeted ~2.5-3 years after Stage 1 completion.
6. Core Lithium's **Finniss** mine is guided to FY24 spodumene sales of 90-100kt and production of 80-90kt due mainly to lower recoveries, mine plan adjustments and mining rates, and overall production in FY25 is expected to be below FY24 (though excludes any BP33 contribution).
7. Liontown's **Kathleen Valley** is in construction, with completion and commissioning is targeted for mid-2024 on an expanded project potentially delivering ~580ktpa (GSe) of spodumene concentrate, where we factor in the planned accelerated expansion to 700ktpa of spodumene concentrate in CY27E, with the project funding now de-risked following announced debt facilities and an equity raise.
8. Covalent's (SQM/Wesfarmers) **Mt. Holland** project is progressing, targeting production from FY24 and ramping up to ~380ktpa of spodumene concentrate, with studies to double capacity to ~760ktpa underway (now in GSe).

We find a number of additional projects and technology developers looking to utilise Direct Lithium Extraction (DLE) or support its implementation, with DLE increasingly being considered for the next wave of projects across both brownfield expansions at existing operations and emerging explorers/developers. We have previously outlined the potential challenges around DLE implementation, where our view remains that with the ongoing efforts, DLE could be implemented between 2025-30 in both Chile and Argentina. Separately, the applicability of DLE to other potential resources, such as typically low lithium concentration geothermal/oil field brines, may also support the economic development of lower quality resources longer term

Exhibit 18: Australian lithium projects



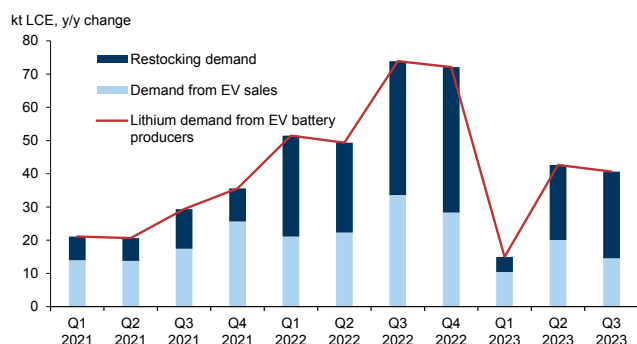
IGO/Tianqi Kwinana Train 2 included above though subject to FID; Covalent Kwinana expansion opportunity being explored; Kemerton Train 3&4 FID takes capacity to 100ktpa.

Source: Company data, data compiled by Goldman Sachs Global Investment Research

Note: The discussion of ex-China supply views are attributable to Hugo Nicolaci who is an equity analyst covering Australia Basic Resources. All other views are those of the Commodities research team.

4. Lithium: All eyes on Chinese supply. Since 2Q2022, we have called for a bear market in lithium prices on a surge that will outpace demand trends. Whilst there have been brief rallies in the lithium market (China 2H2022 EV stimulus and 2Q2023 reopening boost), onshore carbonate prices have continued to trend lower since 4Q2022 and have even fallen 17% lower than our 3Q2023 expected price. This path lower in price has reflected the softening in fundamentals from a slowdown in Chinese EV sales growth volumes (+2.7Mil units in 2022 vs +1.5Mil units in 2023E) alongside aggressive supply growth this year (+246kt LCE GSe). The critical question at the current juncture regards whether supply adjustments generated by the current price level are enough to limit oversupply. It is true that the elevated cost of production for non-integrated lithium carbonate (~\$19k/t and \$27k/t using spot lepidolite and spodumene respectively) puts c.80kty capacity currently in a loss making position ([Exhibit 21](#)). However, the non-integrated cost of production using purchased spodumene (c.120kty in 2024) is expected to fall on average to a cost of \$14,000/t LCE on lagged correction in spodumene prices by mid-next year ([Exhibit 22](#)), on our estimates. Moreover, our [China team](#) has maintained their estimate of total cost for the [integrated lepidolite](#) between \$11-14k/t LCE, which is substantially below current price levels. We continue to see China lithium carbonate prices declining to \$15,000/t LCE on a 12m horizon but lower Asia CJK CIF hydroxide to \$16,500/t (from \$19,000/t previously). We have also lowered our FY averages for lithium prices as shown in [Exhibit 25](#) to account for the realised correction in prices YTD. The main risk, in our view, to a low-cycle in lithium pricing in 2H2024 comes from delays to Chinese expansions from [land quota limitations](#) which is required for [tailing management](#) or slower spodumene ramp-ups.

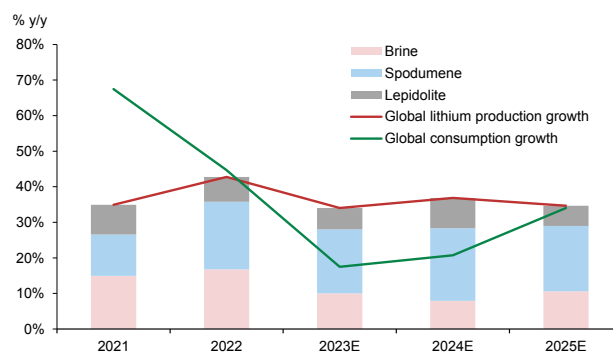
Exhibit 19: Slowing lithium demand growth reflects moderating incremental EV output volumes in China (+1.5mil units YTD vs +2.5mil units in 2022)



Source: Goldman Sachs Global Investment Research, Wind, SMM

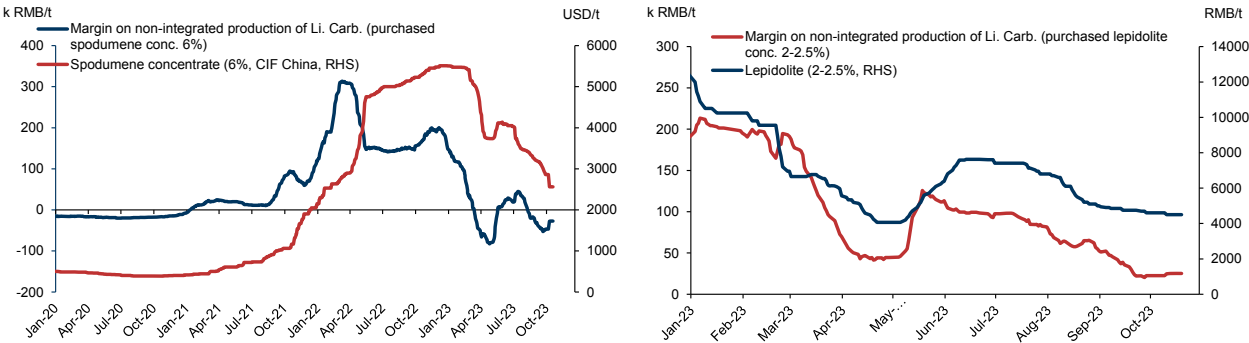
Exhibit 20: A phase of strong supply growth has emerged just as demand growth slows down globally

Global lithium production growth (pre-disruption adjustment)



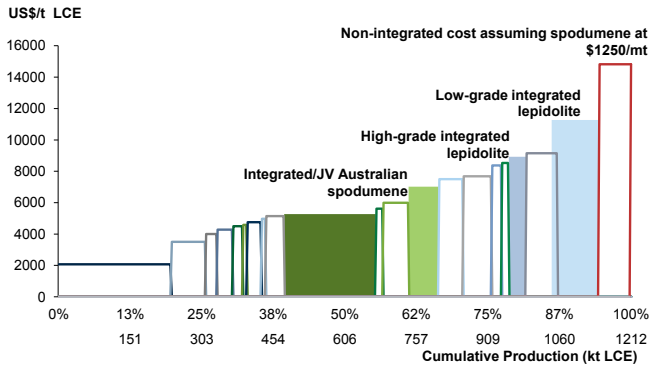
Source: Goldman Sachs Global Investment Research, Company data, Woodmac, BNEF

Exhibit 21: Elevated cost of production for non-integrated lithium carbonate (~\$19k/t and \$27k/t using spot lepidolite and spodumene respectively) puts c.80kty capacity currently in a loss-making position



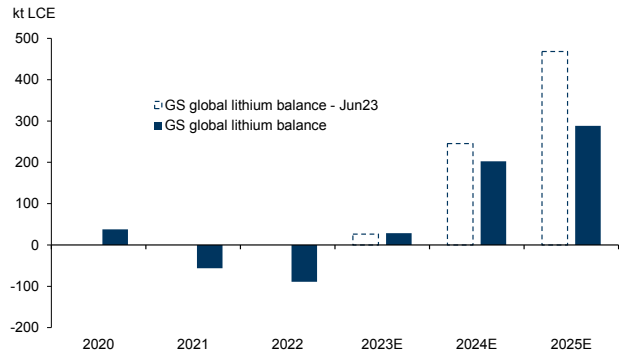
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 22: We expect correction in spodumene prices to lower non-integrated cost of production in the coming quarters
2024 cost curve, excluding royalties and VAT



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research, Woodmac

Exhibit 23: Prices are still substantially above the levels required to balance the market



Source: Goldman Sachs Global Investment Research, Woodmac, SMM, Data compiled by Goldman Sachs Global Investment Research

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Exhibit 24: GS Lithium supply-demand balance

Li ('000 tonnes LCE)	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Global demand											
Consumption - batteries	191	346	538	766	1007	1327	1622	1898	2279	2654	3090
% change y/y	35%	81%	55%	43%	31%	32%	22%	17%	20%	16%	16%
EV	103	230	372	547	732	998	1208	1428	1701	1996	2333
ESS	11	18	34	63	91	119	177	196	252	274	332
E-buses, two-wheeler EVs	20	33	48	64	84	102	120	146	187	235	263
Portable electronics	39	41	46	47	48	50	53	56	60	63	67
Other	18	24	37	46	51	57	64	73	79	87	95
Consumption - ex batteries	130	137	141	144	148	151	155	158	162	165	169
% change y/y	-2%	5%	3%	2%	3%	2%	2%	2%	2%	2%	2%
Ceramics	32	34	35	36	37	38	39	40	41	42	43
Glass-ceramics	27	28	29	30	31	31	32	33	34	35	36
Other	71	74	77	78	80	82	83	85	86	88	90
% of global demand		18%	22%	7%	2%	6%	6%	5%	4%	3%	2%
Global demand (excl. restocking)	321	482	679	910	1155	1478	1777	2057	2441	2820	3258
Global demand (incl. restocking)	341	571	827	971	1173	1571	1878	2154	2536	2917	3338
% change y/y	20%	67%	45%	17%	21%	34%	20%	15%	18%	15%	14%
Global Refined Supply	375	506	752	1099	1595	2149	2561	2900	3142	3235	3287
Brine	172	228	326	438	557	727	852	968	1082	1124	1137
China	36	60	79	114	135	229	271	307	314	320	327
Ex-China	137	168	247	324	422	498	582	661	768	804	810
Spodumene	189	232	342	521	790	1083	1346	1549	1649	1672	1683
China	10	7	14	30	59	86	116	172	188	205	221
Ex-China	179	226	328	491	731	997	1230	1377	1461	1468	1461
Other (Lepidolite or Clay)	13	45	83	140	248	339	363	382	410	439	467
China	13	45	83	140	248	339	363	382	409	435	462
Ex-China	0	0	0	0	0	0	0	0	2	3	5
World output	375	506	752	1099	1595	2149	2561	2900	3142	3235	3287
% change y/y		35%	49%	46%	45%	35%	19%	13%	8%	3%	2%
<i>Total China supply</i>	<i>59</i>	<i>112</i>	<i>176</i>	<i>284</i>	<i>442</i>	<i>653</i>	<i>749</i>	<i>861</i>	<i>911</i>	<i>960</i>	<i>1010</i>
<i>Total ex-China Supply</i>	<i>316</i>	<i>394</i>	<i>576</i>	<i>815</i>	<i>1153</i>	<i>1495</i>	<i>1812</i>	<i>2038</i>	<i>2231</i>	<i>2275</i>	<i>2277</i>
<i>% China supply as of global</i>	<i>16%</i>	<i>22%</i>	<i>23%</i>	<i>26%</i>	<i>28%</i>	<i>30%</i>	<i>29%</i>	<i>30%</i>	<i>29%</i>	<i>30%</i>	<i>31%</i>
Probable supply						32	132	352	584	699	769
Total output (adj. for disruption)	375	506	722	967	1324	1783	2126	2407	2608	2685	2728
% change y/y		35%	43%	34%	37%	35%	19%	13%	8%	3%	2%
Battery Scrap Supply	4	9	16	33	51	76	95	98	119	205	312
% change y/y		2%	2%	3%	4%	4%	4%	4%	5%	8%	11%
Global Balance	38	-56	-89	29	202	288	343	350	190	-27	-298
as % of global supply	10%	-11%	-12%	3%	15%	16%	16%	15%	7%	-1%	-11%

Source: Goldman Sachs Global Investment Research, Company data, BNEF, IEA, Woodmac

Exhibit 25: GS battery-grade lithium price forecasts

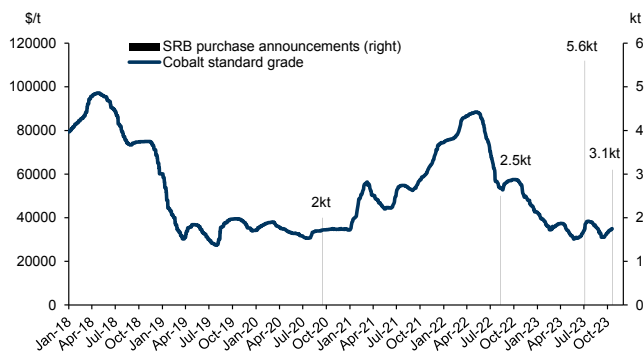
Annual average prices		Lithium carbonate China						Lithium hydroxide China					
		OLD	China lithium carbonate (RMB/t)	OLD	China Lithium carbonate (USD/t, with VAT)	OLD	China Lithium carbonate (USD/t, excluding VAT)	OLD	China lithium hydroxide (RMB/t)	OLD	China lithium hydroxide (USD/t, with VAT)	OLD	China lithium hydroxide (USD/t, excluding VAT)
2019	31/12/2019		70246		10197		9024		84461		12274		10862
2020	31/12/2020		44382		6414		5676		52228		7541		6674
2021	31/12/2021		112974		17535		15518		106738		16567		14661
2022	31/12/2022		465537		69301		61329		450056		66905		59208
2023E	31/12/2023	316872	280733	45327	39971	40113	35372	337639	287935	48296	41067	42740	36342
2024E	31/12/2024	118750	106750	17324	15116	15331	13377	133750	113750	19518	16117	17273	14263

		Asia CIF CJK				Spodumene	
		OLD	Asia lithium hydroxide CIF CJK (\$/t)	OLD	Asia lithium carbonate CIF CJK (\$/t)	OLD	Spodumene 6%, USD/t
2019	31/12/2019		14158		11660		624
2020	31/12/2020		9734		7750		435
2021	31/12/2021		16119		15139		869
2022	31/12/2022		70435		69011		4204
2023E	31/12/2023	54502	46995	48500	42528	4366	3911
2024E	31/12/2024	22074	16491	18324	15741	1763	1250

Source: Goldman Sachs Global Investment Research, SMM, Fastmarkets

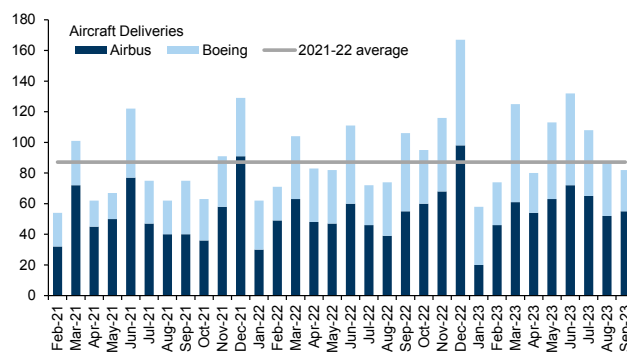
5. Cobalt: Waiting on Chinese capacity. Over the past six months, cobalt prices have remained rangebound between \$13.7-\$17.5/lb on account of relatively modest battery demand, intermediate oversupply, and stockpiling activities. Supply disruptions, news of strategic buying and possible supply rationing have recently boosted prices, and now it is reported that China's SRB is looking to buy more metal after reportedly buying some metal earlier this year ([Exhibit 26](#)). As a result, we think prices will get a boost near-term, but note that this will be an opportunity to sell given the supply glut the cobalt market is facing as Indonesian intermediate and Chinese refining ramps up and CMOC stockpiles are released in the coming months. Furthermore, on the demand side, EV battery demand has been modest due to increasing share of LFP and low-cobalt NMC batteries and consumer electronics demand has remained weak with Chinese electronics output well below the 2021-22 average ([Exhibit 28](#)). Contrary to other sectors, the aerospace sector has remained strong with deliveries resilient and is expected to be a source of strong demand given solid new orders ([Exhibit 27](#)). Overall, we keep our 2023 balance broadly unchanged and continue to see the market in a surplus for the coming years. The key risks to supply comes from a high concentration of both smelter (in DRC) and refined production (in China). We increase our 2023 average price to reflect the recent price rally and 12m price target to \$28,000/t (or \$12.7/lb) (vs \$25,000/t previously) on higher price support for metal prices from producer rationing and restocking activity. Our new FY average forecast for the 2023 cobalt metal price is \$34,750 (or \$15.8/lb), from \$32,000/t previously. Finally, whilst the metal outlook has become relatively tighter, we expect hydroxide prices to remain under pressure from rising MHP production in Indonesia which should also weigh on metal prices over 2024-25 as incremental Chinese refining capacity is brought online. In this context, we keep our 2024-25 cobalt metal average price forecast at \$30,000/t (or \$13.61/lb).

Exhibit 26: News of SRB buying has led to the cobalt metal price moving higher recently



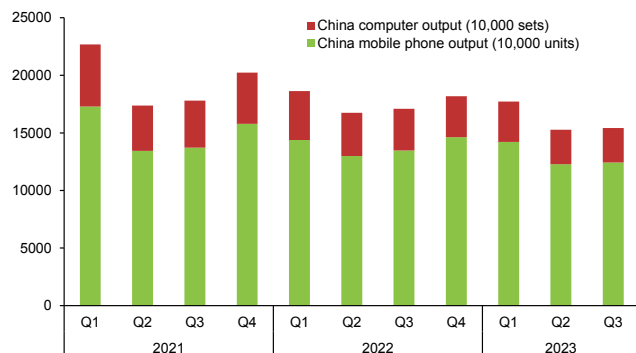
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 27: Aerospace sector has remained strong on resilient deliveries and is expected to be a source of strong demand given solid new orders



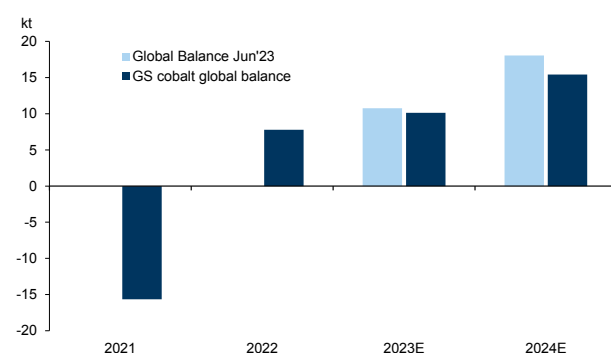
Source: Goldman Sachs Global Investment Research, Airbus, Boeing, Cirium

Exhibit 28: Consumer electronics demand has remained weak with Chinese electronics output well below the 2021-22 average



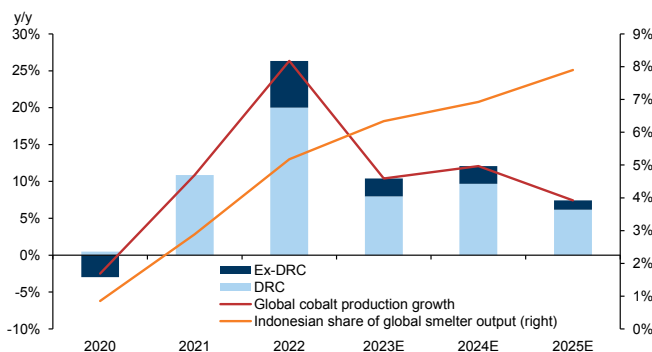
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 29: Cobalt market remains structurally bearish on oversupply from DRC and Indonesia



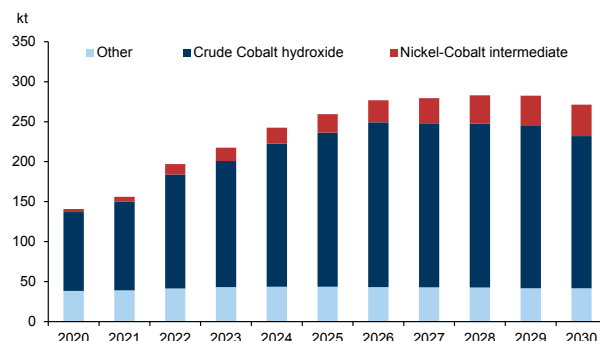
Source: Woodmac, BNEF, Fastmarkets, Goldman Sachs Global Investment Research

Exhibit 30: Rising MHP production in Indonesia should add metal volumes over 2024-25 as incremental Chinese refining capacity is brought online



Source: Goldman Sachs Global Investment Research, Company data, SMM

Exhibit 31: Nickel-Co intermediate based production will be the main source of supply growth from 2026



Source: Goldman Sachs Global Investment Research, Woodmac

Exhibit 32: GS Cobalt supply-demand balance

('000 tonnes)	2020	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Global demand											
Consumption - batteries	83	109	115	133	150	174	194	212	230	256	278
% of total	59%	64%	64%	67%	68%	70%	72%	73%	75%	76%	77%
% change y/y	9%	32%	6%	16%	12%	16%	12%	10%	8%	11%	9%
EV	20	43	50	65	77	98	112	125	135	155	169
ESS	2	3	3	5	6	8	11	11	14	16	17
E-buses, two-wheeler EVs	2	3	2	3	5	5	6	7	8	8	10
Portable electronics	56	57	56	56	57	59	61	64	67	71	75
Other	3	4	4	4	4	4	5	5	5	6	7
Consumption - ex batteries	58	62	65	66	69	73	75	77	79	80	82
% change y/y	-7%	8%	4%	2%	5%	5%	3%	3%	2%	2%	2%
Superalloys	17	18	19	20	21	22	23	23	24	24	25
Tool materials	11	12	12	13	13	13	14	14	14	14	14
Other	30	32	33	33	35	37	39	40	41	42	42
Global consumption	140	171	180	199	219	246	269	289	308	336	359
% change y/y	1%	22%	5%	11%	10%	12%	9%	8%	7%	9%	7%
Global Production											
World smelter output	141	156	197	218	244	262	288	293	300	303	293
% change y/y	-2%	11%	26%	10%	12%	7%	10%	2.0%	2.4%	0.8%	-3.2%
DRC	96	112	143	159	180	195	214	214	215	214	201
% change y/y	1%	16%	28%	11%	13%	8%	10%	0%	1%	-1%	-6%
Ex- DRC	44	44	54	59	64	67	74	80	85	89	92
World refined output	136	150	190	210	235	253	277	283	290	292	282
World output (adj. for disruption)	136	150	182	203	226	243	266	274	281	283	274
% change y/y		11%	21%	11%	12%	7%	10%	3%	2%	1%	-3%
Battery Scrap Supply	4	5	5	7	9	11	13	14	15	22	26
% change y/y		3%	3%	3%	4%	5%	5%	5%	5%	8%	10%
Global Balance	-1	-16	8	10	15	7	11	-1	-13	-31	-59
as % of global supply	-1%	-10%	4%	5%	7%	3%	4%	0%	-5%	-11%	-21%
Cash Prices (annual average)											
Current dollar (\$/t)	33947	51119	69100	34750	30000	30000	44000	44000	44000	44000	44000
Current dollar (\$/lb)	15.4	23.2	31.3	15.8	13.6	13.6	20.0	20.0	20.0	20.0	20.0

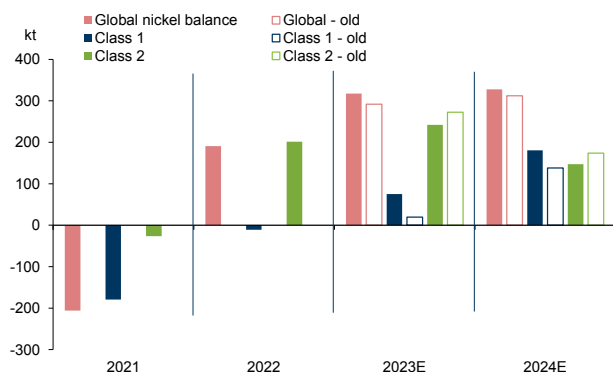
Prices refer to Fastmarket cobalt metal standard grade

Source: Goldman Sachs Global Investment Research, Woodmac, BNEF

6. Nickel's class 1 oversupply is only just beginning. As we highlighted [back in August](#), evidence of surplus pressures have started to mount in the nickel market over the course of 2H this year. Whilst industry feedback has suggested an increasing onshore surplus of class 1 nickel (cathode) from mid-year, it is only since August that Chinese visible stocks have now started build, rising 10kt to now 16kt. These inventory builds have been coupled with deterioration in China arb amidst evidence of rising onshore refined production (+36% y/y ytd). In the ex-China nickel market, whilst LME stock's have remained at low levels, significant pressure on premiums in Europe in particular has pointed to oversupply in the region. These fundamental signals align with our latest supply-demand estimate that places the global nickel market in 318kt surplus this year (vs. previous 292kt surplus), balanced between a 242kt surplus in class 2 nickel and 75kt surplus in class 1 nickel ([Exhibit 33](#)). We expect surplus pressures to mount into 2024 (GSe 328kt surplus), with a significant increase in class 1 surplus (GSe 180kt surplus) driven by an increase in cathode production in both Indonesia and China. However, with LME nickel prices now trading into the cost curve, the probability of supply rationing has increased although there is no evidence yet of this effect with prices close to \$18,000/t. Whilst supply shocks and macro policy shifts can lead to

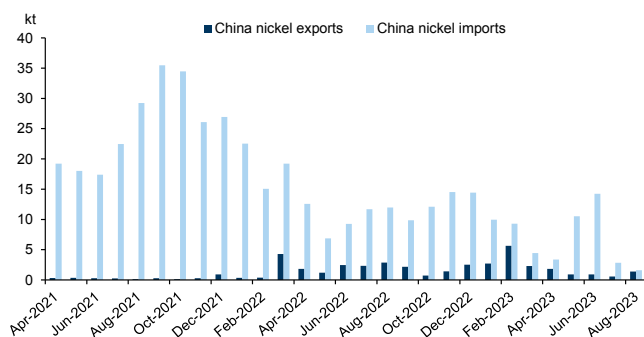
short-covering, given the magnitude of class 1 surplus that we believe still lies ahead in 2024, we remain bearish on nickel (12m target at \$16,000/t). The history of the nickel market suggests a substantial and sustained price overshoot into the cost curve is likely necessary to materially rebalance the forward balance path.

Exhibit 33: Nickel market to remain in a surplus into 2023 as supply continues to grow



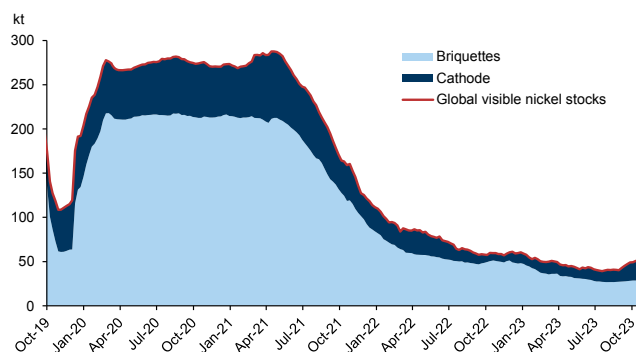
Source: Goldman Sachs Global Investment Research, INSG

Exhibit 35: Chinese import appetite has fallen in recent months



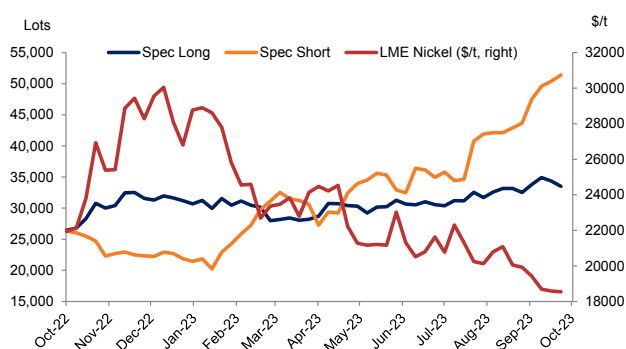
Source: Goldman Sachs Global Investment Research, SMM

Exhibit 37: Sellers remain at risk from low refined metal stocks, which will be key for the next leg lower in prices



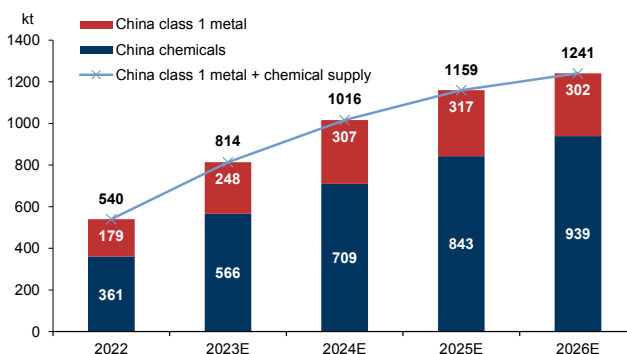
Source: Goldman Sachs Global Investment Research, Bloomberg, SMM

Exhibit 34: Supply shocks can lead to short-covering led price correction as refined metal stocks remain low



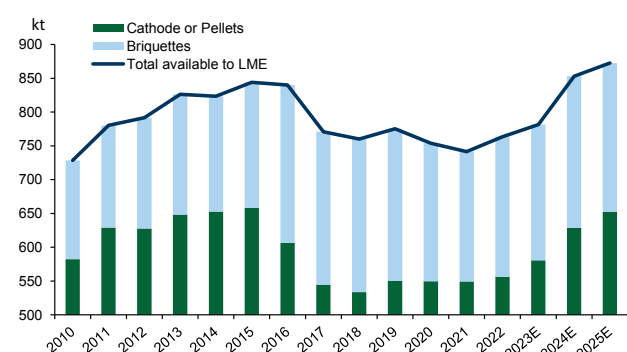
Source: Goldman Sachs Global Investment Research, Bloomberg

Exhibit 36: Chinese refined assets should add c.250kty of metal and nickel sulphate supply each year over 2023-26E



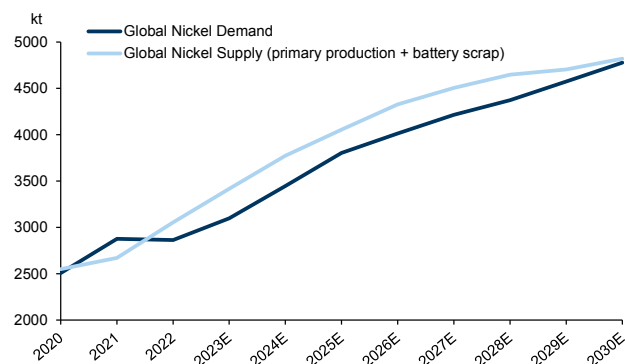
Source: Goldman Sachs Global Investment Research, Woodmac, Bloomberg, Company data

Exhibit 38: Increasing Chinese metal supply should lead to more LME deliverable material in the coming years



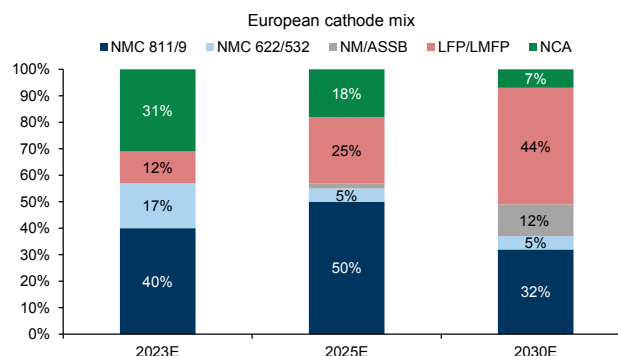
Source: Goldman Sachs Global Investment Research, Woodmac

Exhibit 39: Significant investment in Indonesian nickel projects and ramp-up of class 2 conversion capacity has created a decade of glut



Source: Goldman Sachs Global Investment Research, INSG

Exhibit 40: Rising share of LFP/LMFP batteries is weighing on long-term EV nickel demand



Source: Goldman Sachs Global Investment Research

Exhibit 41: GS Global Nickel SD Model

('000 tonnes)	2020	2021	2022	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Consumption - DM											
US	130	138	134	149	169	222	298	348	386	415	456
% change y/y	-13%	6%	-3%	11%	14%	31%	34%	17%	11%	8%	10%
Europe	290	330	313	327	358	395	436	473	517	558	593
% change y/y	-9%	14%	-5%	4%	10%	11%	10%	9%	9%	8%	6%
Japan	131	158	153	158	166	175	180	187	198	209	217
% change y/y	-15%	21%	-3%	4%	5%	6%	2%	4%	6%	6%	3%
Other DM	7	8	9	9	9	9	9	9	9	10	10
% change y/y	-11%	7%	8%	2%	2%	2%	2%	2%	2%	2%	2%
Sub- DM	559	634	608	642	702	802	923	1018	1110	1192	1275
% change y/y	-9%	13%	-4%	6%	9%	14%	15%	10%	9%	7%	7%
Consumption - EM											
China	1447	1522	1598	1763	1944	2116	2156	2239	2275	2362	2451
Indonesia	206	383	370	372	461	521	554	557	560	563	567
China & Indonesia	1653	1906	1968	2135	2405	2637	2710	2796	2835	2925	3018
% change y/y	10%	15%	3%	8%	13%	10%	3%	3%	1%	3%	3%
Other EM	295	337	288	320	340	365	379	401	428	457	485
% change y/y	-18%	14%	-15%	11%	6%	7%	4%	6%	7%	7%	6%
Sub- EM	1949	2243	2256	2456	2745	3001	3089	3197	3264	3382	3503
% change y/y	5%	15%	1%	9%	12%	9%	3%	3%	2%	4%	4%
Global Consumption	2507	2877	2864	3098	3446	3804	4013	4215	4373	4574	4778
% change y/y	1%	15%	0%	8%	11%	10%	5%	5%	4%	5%	4%
World ex-China & Indonesia	854	971	896	963	1041	1167	1302	1419	1538	1649	1760
% change y/y	-12%	14%	-8%	7%	8%	12%	12%	9%	8%	7%	7%
Global Production											
Global Primary Production	2546	2664	3046	3397	3747	4012	4274	4446	4572	4567	4639
% change y/y	7%	5%	14%	12%	10%	7%	7%	4%	3%	0%	2%
China	760	729	848	1047	1208	1330	1457	1524	1613	1613	1686
% change y/y	-4%	-4%	16%	23%	15%	10%	10%	5%	6%	0%	5%
Indonesia	639	889	1142	1322	1425	1519	1602	1700	1731	1731	1731
% change y/y	65%	39%	28%	16%	8%	7%	5%	6%	2%	0%	0%
ex-China & Indonesia	1147	1046	1056	1028	1114	1162	1215	1222	1229	1223	1223
% change y/y	-4%	-9%	1%	-3%	8%	4%	5%	1%	1%	0%	0%
Global Battery Scrap Supply	3	6	8	19	27	42	54	60	75	138	181
Global Balance	42	-206	191	318	328	250	315	291	274	130	43
Class 1 Nickel balance	-9	-179	-11	75	180	229	253	186	111	-5	-46
Class 2 Nickel balance	52	-27	202	242	147	21	63	106	163	136	90
Cash Prices (annual average)											
Current dollars (\$/t)	13803	18474	28316	20525	18000	18000	-	-	-	-	-
Current dollars (c/lb)	626	838	1284	931	816	816	-	-	-	-	-

Source: Goldman Sachs Global Investment Research, Woodmac, SMM, BNEF

Appendix

Exhibit 42: GS Cobalt Green Demand Model

GS Cobalt Green Demand Model										
000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	9	8	9	10	13	18	23	27	33	37
EV	9	7	8	9	12	17	21	25	30	34
ESS	1	0	1	1	1	2	2	3	3	3
YoY growth	118%	-16%	13%	11%	37%	37%	26%	18%	22%	12%
US	4	4	6	11	21	29	35	38	42	47
EV	4	4	6	10	19	26	32	34	38	42
ESS	0	0	0	1	2	3	3	4	4	4
YoY growth	38%	-3%	50%	74%	84%	39%	22%	7%	11%	11%
China	27	38	50	57	65	69	71	74	82	86
EV	26	35	47	53	60	63	65	67	74	79
ESS	2	2	4	4	5	6	6	7	8	8
YoY growth	122%	39%	34%	14%	14%	6%	2%	5%	11%	6%
RoW	4	4	4	5	6	6	8	10	14	16
EV	4	3	4	5	6	6	7	9	13	15
ESS	0	0	0	0	0	1	1	1	1	1
YoY growth	91%	-16%	19%	26%	10%	2%	23%	35%	36%	16%
Global cobalt green demand	48	55	73	89	111	129	143	157	179	196
EV	43	50	65	77	98	112	125	135	155	169
ESS	3	3	5	6	8	11	11	14	16	17
E-buses, two-wheeler EVs	3	2	3	5	5	6	7	8	8	10
YoY growth	101%	15%	32%	22%	25%	16%	12%	9%	14%	10%

Source: Goldman Sachs Global Investment Research

Exhibit 43: GS Lithium Green Demand Model

GS Lithium Green Demand Model										
'000 tonnes LCE	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	45	51	66	85	134	197	252	323	399	481
EV	42	47	59	75	119	171	221	280	349	419
ESS	3	4	7	10	15	26	31	43	50	62
YoY growth	92%	12%	29%	29%	58%	47%	28%	28%	23%	21%
US	21	28	48	80	168	289	378	449	505	607
EV	20	25	43	71	149	251	331	389	443	530
ESS	2	2	5	9	18	38	47	60	63	78
YoY growth	45%	30%	74%	66%	110%	72%	31%	19%	13%	20%
China	161	303	464	611	758	837	908	1047	1181	1345
EV	149	278	416	544	678	732	801	916	1043	1183
ESS	12	25	48	67	80	105	107	132	138	162
YoY growth	163%	89%	53%	32%	24%	10%	9%	15%	13%	14%
RoW	21	24	32	47	58	63	86	134	185	232
EV	20	22	29	42	51	55	76	116	162	202
ESS	2	2	3	5	6	8	11	18	23	30
YoY growth	47%	13%	35%	45%	23%	9%	37%	55%	38%	26%
Global lithium green demand	281	454	674	907	1220	1505	1770	2140	2504	2928
EV	230	372	547	732	998	1208	1428	1701	1996	2333
ESS	18	34	63	91	119	177	196	252	274	332
E-buses, two-wheeler EVs	33	48	64	84	102	120	146	187	235	263
YoY growth	111%	62%	48%	35%	34%	23%	18%	21%	17%	17%

Source: Goldman Sachs Global Investment Research

Exhibit 44: GS Nickel Green Demand Model

GS Nickel Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	44	40	48	58	88	126	162	202	243	277
EV	36	32	39	49	76	110	142	179	218	251
ESS	2	1	2	2	3	5	5	7	8	9
Wind	6	7	6	7	9	11	14	16	16	17
YoY growth	67%	-8%	18%	23%	51%	43%	29%	25%	20%	14%
US	23	22	33	51	101	168	218	255	285	325
EV	17	17	29	46	94	158	207	242	272	310
ESS	1	1	1	2	4	7	8	10	11	12
Wind	6	4	3	3	4	3	3	3	3	3
YoY growth	20%	-7%	53%	54%	100%	66%	30%	17%	12%	14%
China	152	213	313	389	471	511	561	638	711	771
EV	130	190	274	342	417	452	499	569	637	694
ESS	5	8	12	14	16	19	19	22	23	25
Wind	17	15	27	32	38	39	43	47	50	53
YoY growth	74%	40%	47%	24%	21%	9%	10%	14%	12%	8%
RoW	54	48	64	77	91	98	118	149	181	206
EV	17	15	19	26	32	34	48	73	99	119
ESS	1	1	1	1	1	2	2	3	4	5
Wind	37	32	44	50	58	62	68	74	78	83
YoY growth	-4%	-12%	35%	20%	18%	7%	21%	27%	21%	14%
Global green demand nickel	272	320	453	568	741	888	1040	1223	1395	1551
EV	180	223	316	410	551	677	804	958	1115	1246
ESS	8	11	16	20	24	32	34	43	46	51
E-buses, two-wheeler EVs	18	29	40	47	58	62	74	82	86	99
Wind	65	58	80	92	109	116	128	139	147	155
YoY growth	44%	18%	41%	25%	30%	20%	17%	18%	14%	11%

Source: Goldman Sachs Global Investment Research

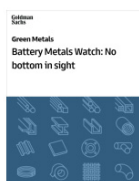
The authors would like to thank Alissa Gorelova for her contributions to this note. Alissa is an intern within the Commodities team.

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20 Jun 2021

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